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ORIGINAL RESEARCH

Sustained Reduction in Cardiopulmonary Fitness in Long COVID

A Report from the RECOVER-adult Cohort Study

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ABSTRACT

BACKGROUND Long-term effect of COVID-19 (Long COVID) may persist for months or years after SARS-CoV-2 infection, but longer-term cardiopulmonary manifestations have not been previously reported.

OBJECTIVES The objective of the study was to characterize cardiopulmonary function after SARS-CoV-2 infection in a digital health substudy of the nationwide Researching COVID-19 to Enhance Recovery Adult Cohort Study.

METHODS Associations between wearable sensor device measures of cardiopulmonary fitness and survey-derived Long COVID symptoms were estimated over a 6-month window at least 6 months after infection using linear regression models adjusted for wear time, age, sex, race/ethnicity, and body mass index.

RESULTS Among 1,475 participants (72% female, 65% non-Hispanic White) a median of 21 months (IQR: 15-31 months) after infection, 498 (34%) had high symptom burden as characterized by the Researching COVID-19 to Enhance Recovery Long COVID Research Index (LCRI). High LCRI (vs low LCRI) was associated with significantly lower heart rate variability (−4.4 ms; 95% CI: −6.5 to −2.4; $P < 0.001$), higher resting heart rate (+1.5 beats/min [+0.7 to +2.4]; $P < 0.001$), fewer metabolic equivalent of task minutes (−96.3 [-128.8 to −63.8]; $P < 0.001$), lower step counts (−1,624 steps/day [-1,952 to −1,296]; $P < 0.001$), and lower activity levels (−7.9 minutes/day very or fairly active [-10.9 to −5.0]; $P < 0.001$). Hierarchal clustering analysis identified two subphenotypes with abnormal cardiovascular measures associated with low quality of life scores.

CONCLUSIONS Long COVID is associated with worse cardiovascular fitness. Additional studies are needed to determine if Long COVID is a novel risk factor for incident cardiovascular disease. (JACC Adv. 2026;5:102923)

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ABBREVIATIONS AND ACRONYMS

BMI = body mass index

DHP = Digital Health Platform

HRV = heart rate variability

Long COVID = long-term effect of COVID-19

LCRI = Long COVID Research Index

MET = metabolic equivalent of task

PROMIS = Patient-Reported Outcomes Measurement Information System

RECOVER = National Institute of Health-funded Researching COVID-19 to Enhance Recovery

Acute cardiopulmonary complications of SARS-CoV-2 infection are mediated by viral infection and replication within cardiac, pulmonary, and vascular tissues with cell surface expression of the angiotensin converting enzyme-2 receptor and concomitant activation of systemic inflammatory pathways. The acute inflammatory response to SARS-CoV-2 infection is known to be associated with pneumonitis, myocarditis, pericarditis, thrombosis, and subacute short-term increases in the incidence of hypertension, arrhythmias, and atherothrombotic and venous thromboembolism events.[1-3]

SARS-CoV-2 infection may also result in various postacute sequelae. Long-term effect of COVID-19 (Long COVID) is defined as an infection-associated chronic condition after SARS-CoV-2 infection and is present for at least 3 months as a continuous, relapsing and remitting, or progressive disease state that affects 1 or more organ systems.[3,4] Long COVID has impacted at least 400 million individuals worldwide with heterogeneous clinical presentations associated with multisystem organ dysfunction, diverse symptomatology, and varying degrees of disability.[5-7] Symptoms traditionally linked to cardiopulmonary disorders, including palpitations, orthostatic intolerance, fatigue, shortness of breath, cough, and chest pain, and postexertional malaise are common in Long COVID, may persist for years, and are often associated with autonomic dysfunction and decreased functional capacity.[8,9]

Wrist-worn wearable sensors ("wearables") passively collect data during daily activities and provide objective quantitative measurements of cardiopulmonary fitness and activity level over time.[10,11] Prior reports of wearables sensor data in post-COVID populations have described short-term changes in cardiopulmonary fitness within 6 months of the inciting COVID infection,[12,13] but long-term effects of SARS-CoV-2 infections on cardiopulmonary symptoms have not been characterized. Accordingly, we conducted a digital health substudy of the nationwide National Institute of Health-funded Researching COVID-19 to Enhance Recovery (RECOVER) Adult Cohort Study to examine the long-term (median of about 2 years) cardiopulmonary consequences of

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Long COVID with wearable sensor-derived cardiopulmonary measures and participant surveys of cardiopulmonary symptoms (**Central Illustration**).

METHODS

OVERVIEW OF THE RECOVER-ADULT COHORT STUDY. Participants were enrolled at 83 participating sites in the United States, including Puerto Rico. The study protocol was posted on the clinicaltrials.gov website (NCT05172024) and approved by the Institutional Review Board at NYU Grossman School of Medicine and collaborating sites. Enrollment in the parent Adult Cohort Study began in 2021, and all participants provided written informed consent.

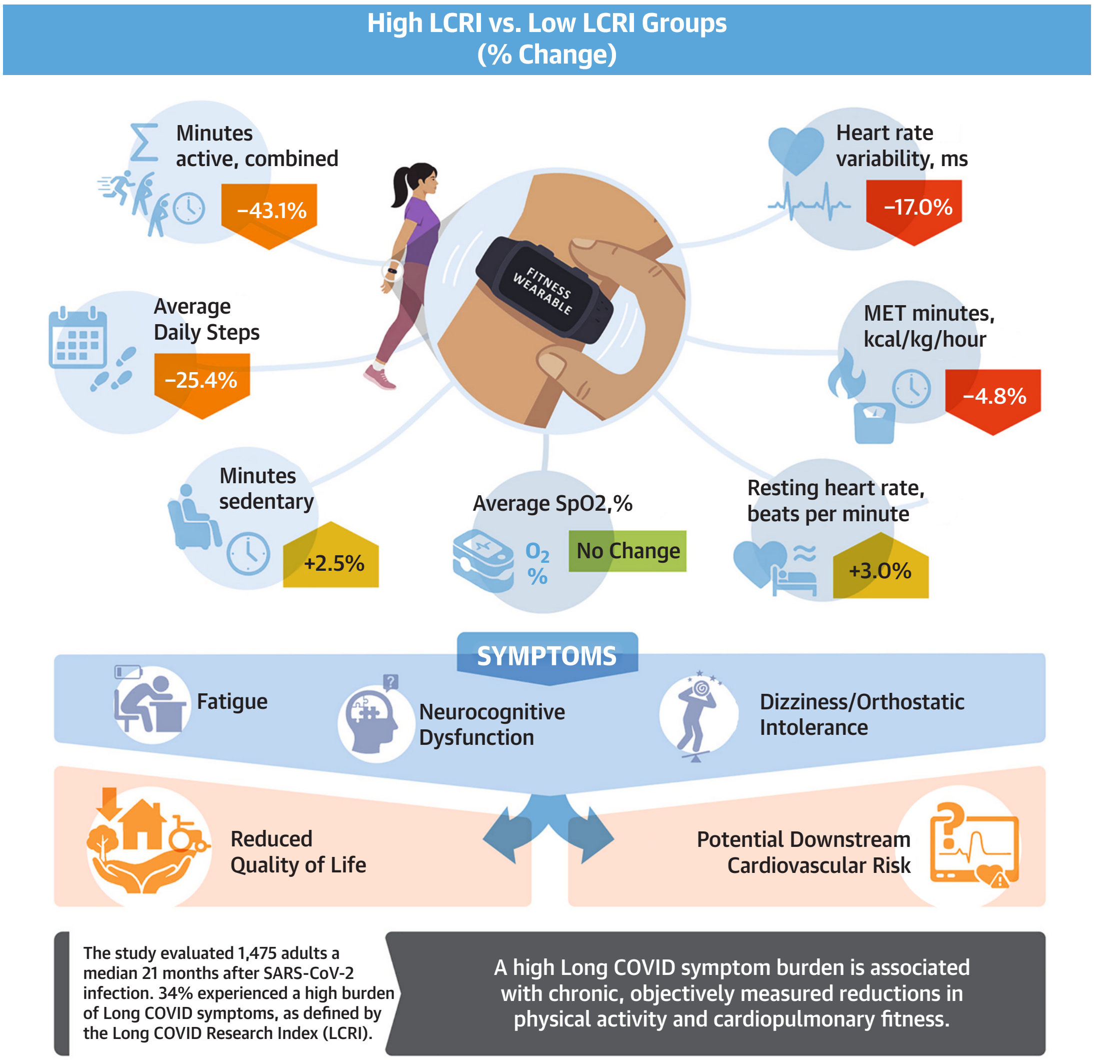
PARTICIPANTS IN THE DIGITAL HEALTH SUBSTUDY. Study sites invited participants enrolled in the RECOVER Adult Cohort Study to join the digital health substudy starting on March 27, 2023. Participants who enrolled in the substudy were asked to connect and share data from their personal consumer-grade wearables or order a study-provided device (Fitbit Sense 2 or Fitbit Charge 5), available to participants at no cost. To include only participants with sufficient time after infection to develop Long COVID, analysis was limited to participants who had a study visit at least 6 months after an index SARS-CoV-2 infection.

A total of 6,529 participants enrolled in the digital health substudy (43% of 15,161 enrolled in the parent RECOVER Adult Cohort Study). Among digital health substudy participants, 4,122 (62%) connected a wearable device. We limited our analysis to Fitbit data since Fitbits comprised the majority (83% of 4,122) of wearables used by cohort participants who connected a wearable to the Digital Health Platform (DHP) substudy. Among the 3,412 participants who shared Fitbit data, 1,596 were excluded because of insufficient evaluable data (defined as at least 5 days of at least 8 hours of data during the 6-month data collection window). This minimum data threshold was instituted to limit bias due to selective sampling times, cyclical patterns of activity, and higher levels of use immediately after starting the use of wearables (Hawthorne effect).

WEARABLES MEASURES. We extracted cardiopulmonary and activity measures from Fitbit devices during a 6-month window centered on the first 2 consecutive symptom survey time points available at least 6 months after the index SARS-CoV-2 infection (Supplemental Table 2 and Supplemental Figure 2): 1) cardiovascular measures (resting heart rate, active heart rate, heart rate variability [HRV] [root-mean-square of the successive normal sinus time between consecutive heart beats difference in milliseconds, ms],[14] metabolic equivalent of task [MET] minutes [1 MET minute is equal to basal metabolic energy expended in 1 minute, 1 kcal/kg/hour],[15] and cardio fitness score [unit-less Fitbit estimate of predicted maximum oxygen uptake (VO_2 max) based on a proprietary algorithm incorporating resting heart rate, age, sex, weight, and other user-entered characteristics])[16]; 2) physical activity (step count, activity level at 4 intensities [minutes sedentary, lightly active, fairly active, or very active])[14-16]; and 3) pulmonary measures during sleep (peripheral oxygen saturation [SpO_2] [average, minimum, and maximum in percent (%)], respiratory rate). Methods for wrist-worn wearable sensor data from Fitbit, derived using accelerometers and photoplethysmography, have been previously described.[17-19] We analyzed Fitbit daily summaries, with the following exceptions: 1) daily MET minutes: sum of minute-level data provided by Fitbit; 2) daily average active heart rate: averaged minute-level heart rates from Fitbit's activity logs; and 3) "combined active minutes:" sum of the daily summary of minutes "very active" plus minutes "fairly active." The mean of daily values over the 6-month window provided a single value for each of the wearable measures per participant.

PARTICIPANT-REPORTED INFORMATION. The RECOVER Adult Cohort Study assessed symptoms and participant-reported diagnoses on questionnaires (Supplemental Table 1) at enrollment and every 3 months.[20] Because symptoms may change over time, we used data from 2 consecutive completed questionnaires that were 1 to 6 months apart and after the start of digital health data collection. We estimated Long COVID symptom burden using the RECOVER 2024 Long COVID Research Index (LCRI) (range 0-30, with LCRI index score ≥11 previously determined as the optimal threshold to identify likely Long COVID in the RECOVER adult cohort).[8] Each symptom was deemed "positive" for the current analysis if a participant reported the symptom at 1 or both time points or "negative" if symptom was absent at both time points. Individuals with LCRI 11 to 30 for at least 1 time point during the 6-month observation period are referred to as "high LCRI." Individuals with LCRI 0 to 10 at both time points were considered "low LCRI." In a sensitivity analysis, we compared high LCRI to LCRI = 0, a group reporting no symptoms in the RECOVER questionnaire. We also examined participant-reported cardiopulmonary symptoms, other common symptoms of Long COVID, symptoms indicating myalgic encephalomyelitis/chronic fatigue syndrome,[21] diagnosis of dysautonomia, and the Patient-Reported Outcomes Measurement Information System

CENTRAL ILLUSTRATION Sustained Reduction in Cardiopulmonary Fitness: A RECOVER Substudy



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In this secondary analysis of 1,475 RECOVER study participants, a cross-sectional analysis was performed using passively collected data from wrist-worn wearable devices. At a median 21 months from index SARS-CoV-2 infection, 34% of those studied had a high burden of Long COVID symptoms, based on the Long COVID Research Index (LCRI). Compared to those with low LCRI, participants with high LCRI had reduced daily step count, less time active and higher time sedentary. In addition, the degree of decreased heart rate variability and total MET minutes are suggestive of increased cardiovascular risk or mortality in other populations. A high Long COVID symptom burden is associated with reduced quality of life and objectively measured reductions in physical activity and cardiopulmonary fitness. Prospective studies will need to determine if these metrics of reduced physical activity and cardiopulmonary fitness translate into increased cardiovascular risk. MET = metabolic equivalent of task; SpO_2 = peripheral oxygen saturation.

(PROMIS) Global Health-10 measure v1.2, a 10-item patient-reported survey to assess participant's overall physical and mental health. The PROMIS Global Physical Health and PROMIS Global Mental Health subscales were examined separately; higher values on the measures indicate better health. PROMIS scores were converted to T-scores with a mean score of 50 and an SD of 10 points.[22,23]

STATISTICAL ANALYSIS. Overview. We used cross-sectional analyses to evaluate associations between wearables measures and Long COVID symptoms collected within a 6-month window (Supplemental Figure 2). All analyses were performed using R (R Foundation for Statistical Computing, 4.4.0).

Association between wearables measures and Long COVID. In the primary analysis, individual wearables measures were modeled in linear regression as the dependent variable against LCRI (high vs low LCRI). For each analysis, 4 models were considered: 1) base model adjusting for average wear time; 2) model 1 plus wear time and age, sex, and race/ethnicity; 3) model 2 plus body mass index (BMI)[24]; and 4) model 3 plus alcohol consumption, smoking, seasonality, time between surveys, and the number of Fitbit records in the time window. Model 3 was considered the primary model, as it is adjusted for potential confounders of cardiopulmonary health and Long COVID. Participants who did not report their sex or reported intersex (0.3% of the digital health substudy) were grouped with the male group for the purpose of analysis. Analyses were repeated comparing high LCRI vs LCRI = 0 groups. To adjust for testing multiple hypotheses, we used the Benjamini-Hochberg method using the significance cutoff of q-value <0.05.[25]

Clustering analysis. Hierarchical clustering was performed using a subset of wearables measures that were strongly correlated with global physical and mental health scores (for external validity). Each measure was normalized by subtracting the mean and dividing the SD. Clusters were generated using "hclust" in the R "stats" package with the Ward.D2 agglomeration method to minimize within-cluster variance. The dendrogram was cut using a hybrid algorithm in the "cutreeDynamic" function in the "dynamicTreeCut" package. The optimal cluster number was determined by visual inspection of the scree plot, selecting the number of clusters at the elbow of the plot (Supplemental Figure 8), and confirmed using Bayesian Information Criterion in the "mclustBIC" function of the "mclust" package (Supplemental Figure 8). We further investigated the demographics, LCRI status, and distribution of overall health within each cluster.

RESULTS

PARTICIPANT DEMOGRAPHICS AND CLINICAL CHARACTERISTICS. Most of the 1,475 participants in the digital health substudy were females (76%), non-Hispanic White (65%), overweight or obese (70%), and had low LCRI (66%); **Table 1**. The digital health substudy participants resemble the overall RECOVER-Adult Cohort Study, except the digital health subgroup included 4% fewer Hispanic and 5% fewer non-Hispanic Black participants. Digital health subgroup participants with high LCRI (vs low LCRI) were 3 years older on average and more frequently overweight or obese (77% vs 64%).

ASSOCIATION OF CARDIOPULMONARY WEARABLES MEASURES WITH LONG COVID. Participants had a median of 4 months of wearables data with 21 hours of wear time per day (Supplemental Table 3). The first time point for participant-reported information was a median of 21 months (IQR: 15, 31) for all participants included in this report, 27 months (IQR: 18, 36) after SARS-CoV-2 infection in the high LCRI group, and 18 months (IQR: 12, 27) in the low LCRI group. The proportion of participants with missing participant-reported information at time point 2 and months between time points 1 and 2 was similar in the high and low LCRI groups.

High LCRI was associated with lower HRV, elevated resting heart rate, lower average step count, and reduced active minutes, when compared with low LCRI (**Figure 1**, **Table 2**). SpO_2 and respiratory rate during sleep were similar between groups. A comparable pattern was noted in analyses stratified by sex, age, and BMI (Supplemental Table 4-7). After adjusting for wear time, age, sex, race and ethnicity, and BMI in a regression model, high LCRI (vs low LCRI) was associated with significantly higher resting heart rate and less HRV in the high LCRI group. Adjusted differences also indicate fewer steps per day, minutes of activity, and MET minutes; as well as more minutes sedentary. (**Figure 2**, Supplemental Table 8). Group differences in mean SpO_2 (−0.2%) and maximum SpO_2 (−0.3%) during sleep were small but statistically significant. No between-group differences in minimum SpO_2 or respiratory rate during sleep were observed. A comparable pattern was present comparing high LCRI vs low LCRI regardless of the model employed in the regression analyses (Supplemental Figures 3 to 5, Supplemental Table 9). We also observed comparable, but numerically larger differences in a sensitivity analysis restricting the comparison groups to high LCRI and LCRI = 0 (Supplemental Figure 6, Supplemental Table 10). For example, after model 3 adjustments, high LCRI was associated with 2,162 fewer steps per day (95% CI: −2,576 to −1,748) vs LCRI = 0, compared to 1,624 fewer steps per day (95% CI: −1,952 to −1,296) (Supplemental Table 8) vs low LCRI.

ASSOCIATION OF SYMPTOMS AND CONDITIONS WITH DIGITAL CARDIOPULMONARY MEASURES. Analyses of partial correlations indicate that symptoms, participant-reported conditions, and worse PROMIS Global Physical Health and PROMIS Global

TABLE 1 Demographic and Body Mass Index of Study Population in the DHP Substudy Compared to the Overall RECOVER Adult Cohort Parent Study

Characteristic	RECOVER Adult Cohort Parent Study (N = 15,161)	DHP Substudy Overall (N = 1,475)	DHP Substudy Low LCRI (n = 977)	DHP Substudy High LCRI (n = 498)
Sex assigned at birth				
Male	4,086 (28%)	354 (24%)	256 (26%)	98 (20%)
Female	10,551 (72%)	1,115 (76%)	716 (74%)	399 (80%)
Intersex	5 (<0.1%)	2 (0.1%)	2 (0.2%)	0 (0.0%)
Age at enrollment, years	45 (34, 59)	45 (35, 58)	44 (34, 58)	47 (37, 57)
Race and ethnicity				
Non-Hispanic White	8,335 (55%)	962 (65%)	624 (64%)	338 (68%)
Hispanic	2,548 (17%)	197 (13%)	129 (13%)	68 (14%)
Non-Hispanic Black	2,154 (14%)	133 (9.0%)	88 (9.0%)	45 (9.0%)
Non-Hispanic Asian	853 (5.6%)	95 (6.4%)	82 (8.4%)	13 (2.6%)
Mixed race/other/missing	1,271 (8.4%)	88 (6.0%)	54 (5.5%)	34 (6.8%)
Body mass index (BMI)[a]				
Underweight (<18.5)	211 (1.5%)	21 (1.4%)	16 (1.6%)	5 (1.0%)
Normal weight (18.5-24.9)	4,117 (29%)	432 (29%)	325 (33%)	107 (22%)
Overweight (25.0-29.9)	4,241 (30%)	419 (29%)	283 (29%)	136 (27%)
Obese (>30.0)	5,691 (40%)	597 (41%)	349 (36%)	248 (50%)

Value are n except for age: median years (quartile 1, quartile 3). Number of individuals in the RECOVER Adult Cohort Study (parent study), DHP substudy overall, DHP substudy high LCRI, and DHP substudy low LCRI groups with missing data for each variable include: sex assigned at birth: 519, 4, 3, and 1, respectively; age at enrollment: 43, 0, 0, and 0, respectively; and Body mass index: 901, 6, 4, and 2; and LCRI: 1,547, 1, 0, and 0, respectively. The proportion of participants with high LCRI at 6 months after index SARS-CoV-2 infection in the RECOVER Adult Cohort Study and DHP substudy are 18% and 26%, respectively. [a]WHO BMI categories.[24]

DHP = Digital Health Platform; LCRI = Long COVID Research Index; RECOVER = National Institute of Health-funded Researching COVID-19 to Enhance Recovery.

Mental Health scores were associated with worse wearables-measured physical health (less physical activity, higher resting heart rate, and lower HRV) as measured by wearables (3). Higher LCRI score, as a continuous measure, was associated with worse wearable-measured health metrics (less physical activity, higher resting heart rate, and lower HRV). This pattern was similar for many individual symptoms; however, associations were less consistently observed for chronic cough, dysautonomia, headache, as well as for wearables measures of SpO_2 and respiratory rate.

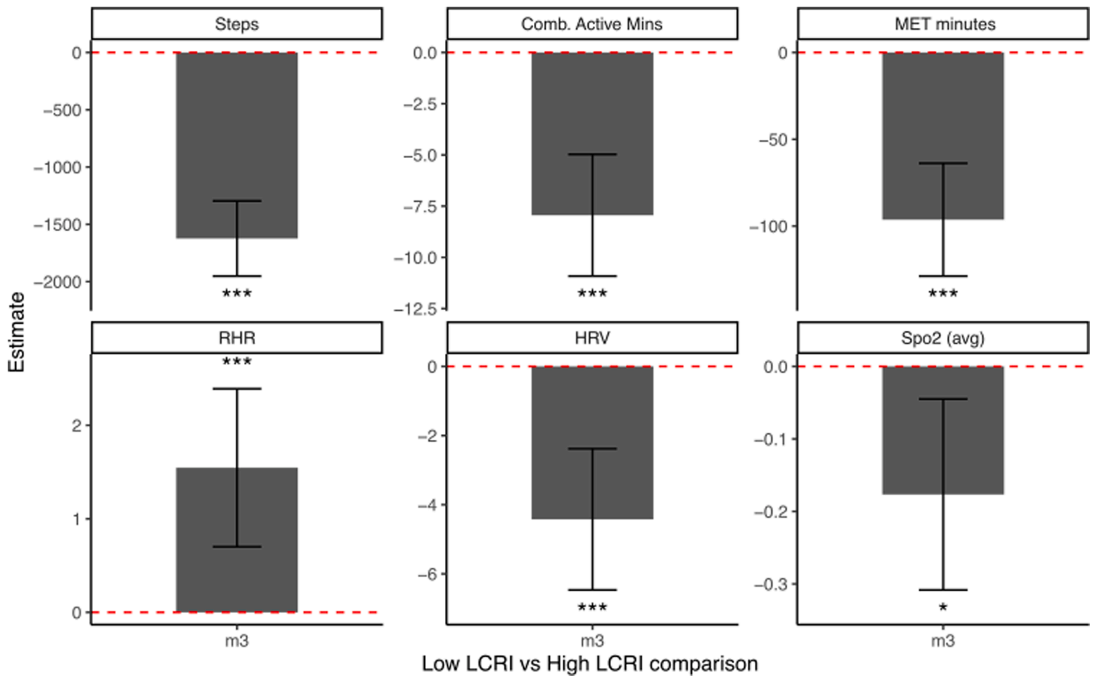
CLUSTER ANALYSIS OF DIGITALLY-DERIVED CARDIOPULMONARY MEASURES. We performed hierarchical clustering using 6 digital measures (HRV, resting heart rate, steps, minutes very active, minutes lightly active, and average SpO_2) to compare wearable data patterns with participant-reported information. We identified 6 primary clusters showing different patterns of digital health measures (Table 3, Figure 3, Supplemental Figure 7). We numbered the digital health clusters (1-6) in descending order of the prevalence of high LCRI (57% to 20%). Notably, cluster 1 ("low physical activity with low SpO_2") and cluster 2 ("low physical activity without low SpO_2"), show the lowest activity profiles, as well as low HRV and high resting heart rate. Both show a high proportion of high LCRI and participants reporting postexertional malaise, along with lower PROMIS Global Physical and Mental Health scores. Relative to cluster 1, cluster 2 tends to be younger (Table 3).

DISCUSSION

In this cross-sectional analysis of about 1,500 adults at a median of 21 months after SARS-CoV-2 infection, higher Long COVID symptom burden was associated with substantial reductions in wearable-derived measures of cardiopulmonary fitness and physical activity. Hierarchical clustering revealed subgroups of participants with distinct profiles of several abnormal cardiopulmonary measures known to be associated with an increased risk of incident cardiovascular events.

A prior study[13] reported reduction in HRV with onset of acute COVID infection and subacute phase (< 6 months). Our study demonstrated a reduction of 4.4 ms group mean difference in HRV between high LCRI and low LCRI at a median of 21 months after SARS-CoV-2 infection. The lowest HRV was seen in clusters 1 and 2 (low physical activity with low SpO_2, and low physical activity without low SpO_2, respectively), which also showed the lowest PROMIS Global Physical and Global Mental Health survey scores,[26,27]

FIGURE 1 Digital Health Measure Differences Between High and Low LCRI



Differences (high LCRI minus low LCRI groups) in step count per day (Steps), very plus fairly active minutes (Comb. Active Mins), MET minutes, resting heart rate (RHR) in beats per minute, heart rate variability (HRV) in ms, and average SpO_2 in % (SpO_2) between groups. Estimates < 0 indicate lower values in the high LCRI group, adjusted for wear time, age, sex, race/ethnicity, and body mass index (model 3 [m3]; see Methods for details). Error bars indicate 95% CIs; 2-tailed significance ***0.001, and *0.05. LCRI = Long COVID Research Index; MET = metabolic equivalent of task; SpO_2 = peripheral oxygen saturation.

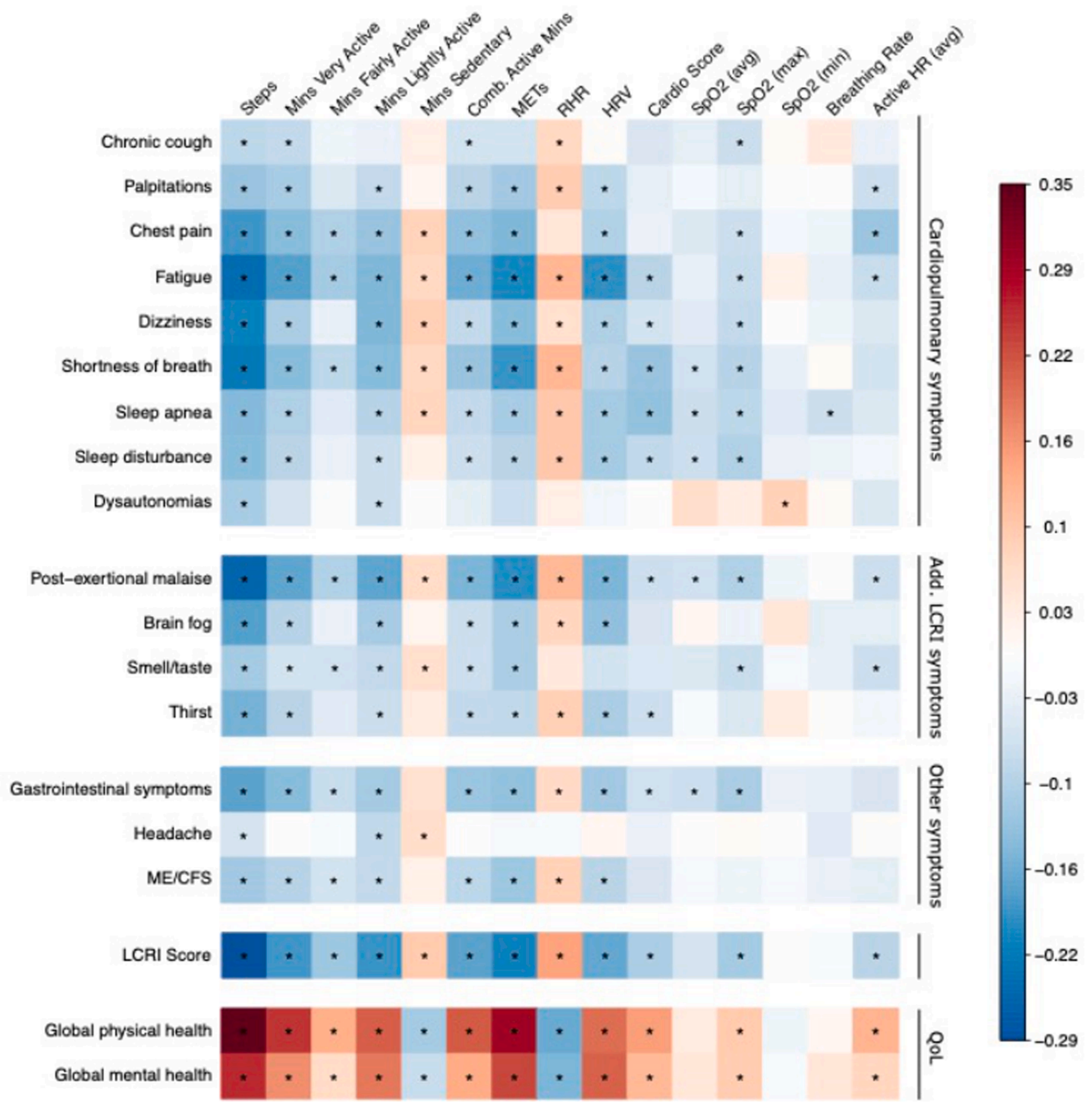
TABLE 2 Wrist-Worn Digitally-Derived Cardiopulmonary Measures in High vs Low LCRI Groups

	Low LCRI (n = 977)	High LCRI (n = 498)
Average daily steps	7,497 (5,677, 9,535)	5,593 (3,867, 7,614)
Minutes very active	12.0 (5.2, 24.0)	5.4 (2.2, 13.2)
Minutes fairly active	13.2 (7.5, 20.5)	8.9 (4.4, 17.0)
Minutes lightly active	223.4 (185.0, 269.6)	201.3 (158.6, 250.5)
Minutes sedentary	799.5 (708.7, 969.9)	819.8 (726.7, 966.4)
Minutes active, combined[a]	26.7 (14.3, 45.4)	15.2 (7.0, 30.2)
MET minutes, kcal/kg/hour	2,212 (2,075, 2,368)	2,105 (1,938, 2,268)
Resting heart rate, beats per minute	67.0 (60.9, 72.5)	69.0 (63.5, 74.4)
Average active heart rate, beats per minute	103.8 (98.1, 109.8)	104.2 (96.8, 110.0)
Heart rate variability, ms	27.7 (20.4, 41.0)	23.0 (17.0, 33.8)
Cardio fitness score[b]	37.1 (31.6, 42.6)	33.6 (29.3, 39.1)
Average SpO_2, %	95 (94, 96)	95 (94, 96)
Maximum SpO_2, %	98 (97, 98)	97 (96, 98)
Minimum SpO_2, %	92 (92, 93)	92 (91, 93)
Respiratory rate, breaths per minute	15 (14, 17)	15 (14, 17)
First questionnaire, months after infection (timepoint 1)	18 (12, 27)	27 (18, 36)

Values are reported as median (Q1, Q3), unadjusted for wear time, age, sex, race/ethnicity, and body mass index. [a]Minutes active is the sum of daily minutes very active and minutes fairly active. [b]Cardio fitness score is a Fitbit proprietary algorithm estimating VO_2 max (see Methods).

MET = metabolic equivalent of task (1 MET = 3.5 mL/kg/min O_2 consumption = 1 kcal/kg/min) minutes; SpO_2 = peripheral oxygen saturation; other abbreviation as in Table 1.

FIGURE 2 Associations Between Digital Health Measures, Symptoms, and Conditions



Partial correlations between digital health measures and symptoms, conditions, and PROMIS Global Physical and Global Mental health subscales, adjusted for wear time, age, sex, race/ethnicity, and BMI. Red color indicates a positive partial correlation (higher value of one measure is associated with higher value of the other), whereas blue indicates a negative partial correlation. An asterisk ('*') indicates statistical significance at FDR 0.05. HR = heart rate; HRV = heart rate variability; ME/CFS = myalgic encephalomyelitis/chronic fatigue syndrome; QoL = quality of life; RHR = resting heart rate; other abbreviations as in Figure 1.

and increased prevalence of postexertional malaise. Postexertional malaise in Long COVID has been previously linked to reduced skeletal muscle mitochondrial density and skeletal muscle necrosis, demonstrating a physiological basis of activity limitations in this population.[28]

Systematic meta-analyses of general population studies have established associations between higher mortality and both lower HRV and higher resting heart rate.[29,30] The 4.7 ms reduction in HRV in the high vs low LCRI groups is similar to the expected decrease in HRV after 10 years of aging.[14] We also observed an approximate 100 MET minutes reduction in the high (vs low) LCRI group. Results of a large population-based cohort that assessed physical activity using a self-reported questionnaire indicate

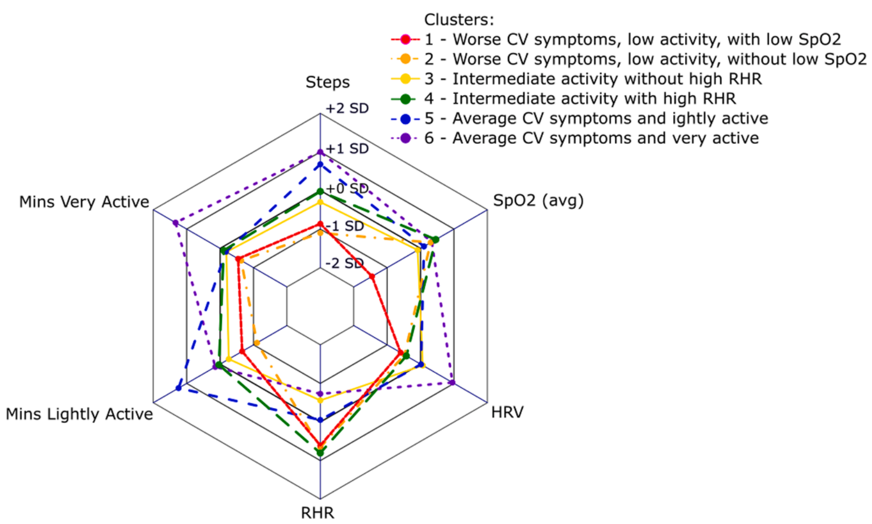
TABLE 3 Cluster Analysis of Digitally-Derived Cardiopulmonary Measures

	Cluster 1 Low Physical Activity and Low SpO_2 (n = 134)	Cluster 2 Low Physical Activity Without Low SpO_2 (n = 165)	Cluster 3 Intermediate Physical Activity Without High RHR (n = 257)	Cluster 4 Intermediate Physical Activity With High RHR (n = 153)	Cluster 5 Lightly Physically Active (n = 243)	Cluster 6 Very Physically Active (n = 213)
Sex assigned at birth						
Female	114 (85%)	133 (81%)	179 (70%)	127 (84%)	199 (82%)	134 (63%)
Male	20 (15%)	32 (19%)	76 (30%)	25 (16%)	44 (18%)	78 (37%)
Unknown	0	0	2	1	0	1
Long COVID category						
High LCRI	76 (57%)	90 (55%)	83 (32%)	47 (31%)	62 (26%)	42 (20%)
Low LCRI	58 (43%)	75 (45%)	174 (68%)	106 (69%)	181 (74%)	171 (80%)
ME/CFS	6 (5.2%)	14 (9.8%)	7 (3.0%)	7 (5.0%)	3 (1.4%)	3 (1.5%)
Unknown	19	22	22	12	27	12
Dysautonomia	1 (0.9%)	14 (9.6%)	10 (4.3%)	7 (5.0%)	4 (1.8%)	4 (2.1%)
Unknown	21	19	22	12	24	18
Postexertional malaise	84 (64%)	107 (66%)	103 (40%)	66 (44%)	80 (34%)	56 (27%)
Unknown	3	4	2	2	5	2
Age at enrollment, years	55 (44, 62)	44 (35, 55)	50 (37, 62)	41 (34, 49)	41 (33, 53)	42 (30, 59)
PROMIS Global Physical, t-score	43 (39, 48)	41 (36, 49)	49 (42, 54)	48 (41, 54)	51 (45, 56)	52 (47, 58)
Unknown	0	0	1	0	0	0
PROMIS Global Mental, t-score	44 (39, 50)	41 (36, 48)	48 (41, 53)	45 (40, 51)	50 (44, 54)	51 (44, 56)
Unknown	0	0	1	0	0	0

Value are n (%), except for age: median years (quartile 1, quartile 3). Results of hierarchical clustering based on 6 digital measures (steps, minutes very active, minutes lightly active, resting heart rate, heart rate variability, and average SpO_2). Six primary clusters were identified, presented in the Table in descending order of the proportion with high LCRI (57% to 20%, cluster 1-6), also corresponds a descending order of the proportion with postexertional malaise (64% to 27%), age (55 years to 42 years), worse PROMIS Global Physical Health score (43-52, lower scores indicate worse health), and worse PROMIS Global Mental Health score (44-51, lower scores indicate worse health).

ME/CFS = myalgic encephalomyelitis/chronic fatigue syndrome; PROMIS = Patient-Reported Outcomes Measurement Information System; RHR = resting heart rate; other abbreviation as in Tables 1 and 2.

FIGURE 3 Radar Plot Showing Clusters and Digital Health Metrics



Plots of clusters derived from digital health data. CV = cardiovascular; other abbreviations as in Figures 1 and 2. Cluster 1 ("low physical activity with low SpO_2") and cluster 2 ("low physical activity without low SpO_2"), show the lowest activity profiles, as well as low heart rate variability and high resting heart rate. Cluster 5 ("lightly physically active") has higher-than-average step count and higher minutes lightly active, average heart rate variability, and resting heart rate. Cluster 6 ("very physically active") has the highest step count, minutes very active, heart rate variability, and lowest resting heart rate. Cluster 3 ("intermediate activity without high resting heart rate") shows moderate levels across all digital health measures, whereas cluster 4 ("intermediate physical activity with high resting heart rate") shows moderate activity, but high resting heart rate and low heart rate variability.

that a 71 MET minute per day activity increase is associated with a 14% decrease in mortality.[32] In addition, we observe increased minutes sedentary in the high LCRI group, which has been associated with higher rates of mortality and cardiovascular disease.[33-35] Further validation of these cardiopulmonary fitness estimates by wearables may have clinical treatment implications, as each 1 MET increase in peak exercise capacity is associated with a 13% reduction in mortality and 15% reduction in cardiovascular disease risk.[36]

Our findings are also concordant with smaller prior studies (39-553 participants) using direct physiological measurements other than wearable devices demonstrating impaired exercise capacity and autonomic dysfunction in post-COVID syndromes.[26,37-42] In the current report, the median daily step count in the high LCRI group (5,593 steps/day) was below the 25th percentile of both the low LCRI group (5,667 steps/day) and a sample of the U.S. population (5,866 steps/day).[43] A recent meta-analysis noted that taking at least 7,000 steps per day is associated with a marked decrease in the risk of mortality and incidence of several illnesses.[44] Population-based studies demonstrate that small reductions in activity levels detected by wearables are associated with long-term risk or increased mortality and incident cardiovascular events.[45,46]

In the Long COVID population, prior studies suggest that reduced cardiopulmonary fitness and decreased physical activity reflects a combination of limited exercise capacity and intentional pacing to reduce the risk of postexertional malaise,[9,47,48] rather than a lack of motivation or interest in exercise.[49] Our wearable sensor data cannot definitively disentangle the effects of physiologic impairment (eg, ventilatory, circulatory, and neuromuscular limitations) vs intentional effort restriction on the cardiopulmonary fitness among study participants. This study's strengths include a well-characterized, multicenter cohort of nearly 1,500 individuals with Long COVID drawn from 83 sites across the United States with wearables data obtained an average of nearly 2 years after SARS-CoV-2 infection. In this report, we integrated wearables metrics and standardized questionnaires (including PROMIS physical and mental health), offering a comprehensive, multidimensional assessment of Long COVID at 6 months or more after SARS-CoV-2 infection. The interdisciplinary approach to developing this report, including collaboration with people with lived experience of Long COVID, aligns with best practices in research.[50,51]

STUDY LIMITATIONS. There are several caveats to consider when interpreting our findings. The requirement for in-person enrollment into RECOVER Adult Cohort Study likely excluded individuals with more severe postexertional malaise who may be housebound or bedbound and unable to travel to study sites.[51,52] Also, participation in the parent cohort study and DHP substudy was based on self-selection (people opting in to join the study), with a predominance of White females, potentially introducing selection bias and limiting generalizability of our findings. Our wearable sensor data demonstrated significant but small between group differences in measures of cardiopulmonary fitness between groups which would not be considered actionable in clinical practice. Further studies are needed to validate the wearable sensor measures against the gold standard clinical testing. There was a shorter delay between the first SARS-CoV-2 infection and the start of participant-reported data collection in the low LCRI vs high LCRI groups (18 and 27 months, respectively). We can only speculate about the reasons for the difference in timing of enrollment in the DHP substudy by LCRI group, which may include differences in the SARS-CoV-2 variant exposure, availability of vaccination, or other time-varying pandemic treatments that may impact the risk of Long COVID.[53]

The cross-sectional analyses establish an association between Long COVID and objective measures of physical activity and cardiovascular fitness, but do not establish causation. Sufficient data from wearables before SARS-CoV-2 infection were not available in our cohort, so we are unable to quantify individual changes after contracting Long COVID and cannot exclude reverse causation; although reverse causation would not diminish risk conferred by chronic reductions in activity.[26,27] We relied on patient-reported diagnoses of dysautonomia and symptoms of ME/CFS,[21] so underdiagnosis and misclassification are possible.[31,54-56] Consumer wearables use proprietary algorithms developed in healthy populations that may introduce bias when applied to disease populations. Additional studies are needed to validate consumer wearables vs research-grade wearables (e.g., ActiGraph GT3X + hip worn tri-directional accelerometry) and the gold standard clinical testing in a subset of participants with Long COVID.[57] Fitbit devices provided summary reports of SpO_2 and respiratory rate only during sleep, so we were unable to assess the relationships of SpO_2 and respiratory rate with physical activity.

FUTURE DIRECTIONS

Taken together, our findings support the potential for wearables to complement symptom-based assessments in clinical and research settings for Long COVID. Longitudinal analyses of physical activity with long-term follow-up are needed to evaluate whether the indicators of cardiovascular fitness and decreases in physical activity translate into an increased risk of cardiovascular disease. Our cluster analysis suggests that combining symptom data with digital health metrics may enable development of a digital biomarker for Long COVID, which might be useful to evaluate unexplained poor functional capacity in clinical practice and also serve as objective endpoints in Long COVID clinical trials. Advances in wearable sensor technology, including miniaturization of sensors and use of artificial intelligence may further improve the accuracy and clinical utility of wearables for Long COVID.[58,59]

In conclusion, the results from this digital health substudy of the RECOVER Adult Cohort Study demonstrate associations between a high burden of persistent Long COVID symptoms and objective indicators of impaired cardiopulmonary fitness. Findings from the cross-sectional study design in the current report do not establish causation. Additional research, including those employing longitudinal designs, is needed to determine if Long COVID at least 6 months after SARS-CoV-2 infection is a novel risk factor for cardiopulmonary disease.

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APPENDIX For supplemental tables and figures, please see the online version of this paper.